

LESSON GUIDE

Technology Plus

Unit 2: Programming

1.2.2 – Hexadecimal

Lesson Overview:

In this lesson, students will explore the hexadecimal number system, learn to convert between hexadecimal and decimal, and apply their knowledge to real-world IT scenarios, including MAC addresses, error codes, and IPv6 addressing.

Students will:

- Understand the structure of hexadecimal.
- Convert between hexadecimal and decimal.
- Recognize hexadecimal in IT applications.

Guiding Question: Why is hexadecimal used in computing, networking, and cybersecurity instead of decimal or binary?

Suggested Grade Levels: 8-12

Technology Needed: Yes

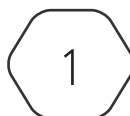
Approximate Time Required: 60-75 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 1.2 – Identify notational systems.
 - Binary
 - **Hexadecimal**
 - **Decimal**
 - Octal

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Lesson Background

Background Information:

Hexadecimal is a fundamental numbering system in computing, bridging the gap between binary, which computers use, and decimal, which humans find intuitive. Since each hexadecimal digit represents four binary bits, it allows for a more compact and readable way to display large binary values without losing precision. This efficiency is why hexadecimal is the preferred format for memory addresses, MAC addresses, IPv6, color codes, and system error messages – all of which IT professionals encounter regularly.

While students may initially focus on conversions between decimal and hexadecimal, professionals rarely perform these calculations manually. Instead, they recognize patterns and apply hex-based knowledge to network troubleshooting, debugging, and system diagnostics. By exposing students to these real-world applications, this lesson ensures they see hexadecimal not just as an abstract math concept but as a critical tool for working with digital systems.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 1.2.2 – Hexadecimal • Activity 1.2.2 – Hexadecimal Seek & Solve • Activity 1.2.2 – Seek & Solve MAC Lookup • Tape
Materials per Student	• Student Notes 1.2.2 – Hexadecimal • Student Handout 1.2.2 – Hexadecimal • CYBER.ORG Range logins (optional) • Clipboard (optional) • Assessment 1.2.2 – Hexadecimal (optional)

Cyber Terms:

Term	Definition
hexadecimal (hex)	a base-16 numeral system using digits 0 through 9 and letters A through F, commonly used as a shorthand for representing binary data

1.2.2 – Hexadecimal

Guiding Question: Why is hexadecimal used in computing, networking, and cybersecurity instead of decimal or binary?

Lesson Launch (5 minutes)

- Distribute Student Handout 1.2.2 – Hexadecimal and encourage students to complete it as you go through today's lesson.
- The objective of slide 3 and its line of questioning is to get students thinking about how precision in color selection relates to the need for a more compact, readable number system in computing.
- When you ask students their favorite color, dig deeper for the students who just give an answer like red, green, blue.
- Ask them about specific shades: brick red, fire engine red, forest green, neon green, Xbox green, navy blue, sky blue?
- The slide includes a color swatch of a very specific purple labeled "My Favorite Color." If you would like to change it, just make sure to update the binary and the hex for the color you select.
- The animations are set as follows:
 - Third bullet, color swatch, and My Favorite Color
 - Fourth and fifth bullet, binary and hex for color swatch

Don't Be Vexed by the Hex (20-25 minutes)

What is Hexadecimal?

- The goal is to help students understand why hexadecimal exists as a bridge between binary and human readability.
- Focus on real-world relevance – hex isn't just an abstract number system; it's used daily in IT fields.
- No animations were used for this section, so pacing is up to the teacher's discretion.

Converting Between Number Systems

- Key Focus: Helping students grasp the step-by-step process for conversions.
- When covering hex-to-decimal conversions, emphasize place value (just like in decimal) and remind students that letters A-F must be pre-converted.
- When covering decimal-to-hex, remind students that the remainders create the hex number (read from bottom to top).
- Walk through the examples slowly – conversions take practice.
- Connect to past learning.

- If students struggled with binary to decimal conversions in the last lesson, highlight how hex follows the same place-value logic but with base-16 instead of base-2.

Why Hexadecimal is Useful in IT

- Key Focus: Hexadecimal makes data more compact, readable, and efficient.
- This section transitions from math-based learning to real-world applications.
- Instead of memorizing theory, students should understand how and why hex is used in IT professions.
- Ask students where they've seen hex before (even if they didn't recognize it as hex at the time – think color codes, IP addresses).
- Display a hex color code and have students compare different shades. You can show them in edit mode of the PowerPoint or by opening another application that lets you choose colors for text and shapes (e.g., MS Word, Google Docs, etc.).
- Reinforce that IT professionals don't constantly convert – they recognize patterns in hex values.

Hexadecimal in Networking

- Key Focus: The importance of hexadecimal in network addressing and device identification.
- Many students won't realize how hex ties into network security, IP addressing, and hardware identifiers.
- The biggest challenge here is making abstract concepts like memory addresses and IPv6 feel real to students.
- Break it down visually – when discussing IPv6, compare 32-bit IPv4 vs. 128-bit IPv6 to show why hex was necessary.
- For MAC addresses, ask: "If I gave you a list of MAC addresses, how could you tell which ones were from Apple or Cisco?"
- Let students practice spotting patterns in manufacturer codes.
- Memory addresses – help students understand that RAM locations use hex because binary would be too long to work with.

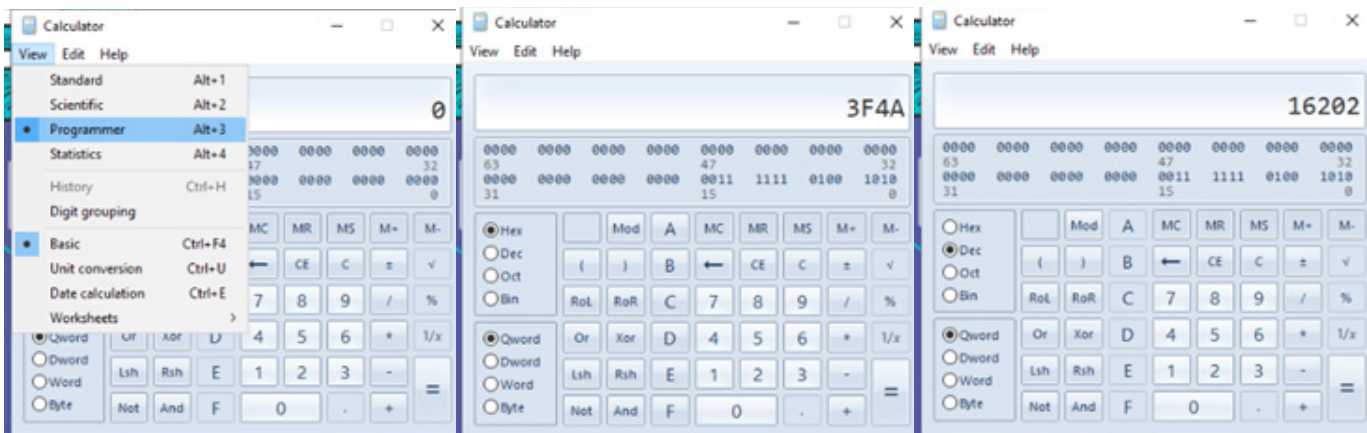
Why IT Professionals Use Hexadecimal

- Key Focus: IT professionals use hexadecimal for efficiency, not just for math.
- Many students assume working in IT means manually converting numbers all the time – but that's not how it works in practice.
- The goal here is to help them understand pattern recognition and practical applications.
- Make it interactive:
 - Have students guess what an error code might mean before looking it up.
 - Show a few MAC addresses and let students guess the manufacturer based on the first three pairs.
- Highlight that mental math isn't always needed:
 - Web designers recognize hex color codes at a glance.

- System admins spot common error codes without converting.

Activity: Hexadecimal Seek & Solve (30-40 minutes)

- This activity works best in a large, open area where students can roam from sheet to sheet. A hallway or your classroom could suffice if you consider personal belonging management before having students up and walking.
- Print the seek-and-solve sheets and use tape to hang them around your space in **alphabetical order**. They are premixed so shuffling is not required.
- Print a copy of the Seek & Solve MAC OUI Lookup document to post as well. Information can vary slightly from site to site and MAC assignments can change over the years, so the reference sheet will make sure that students have the same information that was available at the time this lesson was written.
- Clipboards are included as an optional student supply to provide students with a portable surface to write on, but a sturdy book or personal whiteboard would work great too.
- Print a copy of the answer document to easily check students' work as they navigate the seek and solve. You should be able to quickly scan the order of the letters they are recording after each stop at an answer.
- Make sure that students are only recording letters when they find their answers. A common problem is that students will record the letter from their starting point and throw off their recording or they don't write down letters at all. Either issue adds difficulty to quickly scanning their answers.
- Students will need technology to successfully complete this activity.
- If students are unable to access specific tools due to network restrictions, they can use the CYBER.ORG Range. The built-in Calculator app in Windows Server 22 can be set to Programmer mode for easy hex-to-decimal conversions, and Paint can be used to look up or explore hexadecimal color values.



Putting It All Together (5 minutes)

Discussion Questions:

- What is hexadecimal?

Hexadecimal is a base-16 numeral system that uses the decimal digits 0-9 and A-F to represent the decimal values 0-15.

- Why do you think hexadecimal is used in computing instead of just using binary or decimal?

Hexadecimal acts as a bridge between the binary that computers use and the decimal system that humans find intuitive. It's more compact and readable than binary while still mapping neatly to binary values, making it a practical choice for representing digital data.

- Express your favorite color as a hex code.

Student answers will vary.

- How does recognizing hexadecimal patterns help IT professionals troubleshoot network and systems issues?

Students may offer different answers based on the lesson. Possible responses include:

- The first three hex pairs in a MAC address identify the device's manufacturer, helping with network troubleshooting.
- Memory addresses use hex instead of decimal because it provides a more compact and readable representation of binary.
- Common hex-based error codes help IT professionals quickly diagnose system crashes and failures.
- IPv6 uses hexadecimal instead of decimal-based IPv4, allowing for better scalability and more efficient addressing.

